# Class 11 NCERT Mathematics: Hyperbola

Complete Theory, Formulas, Key Concepts & Interactive Diagram

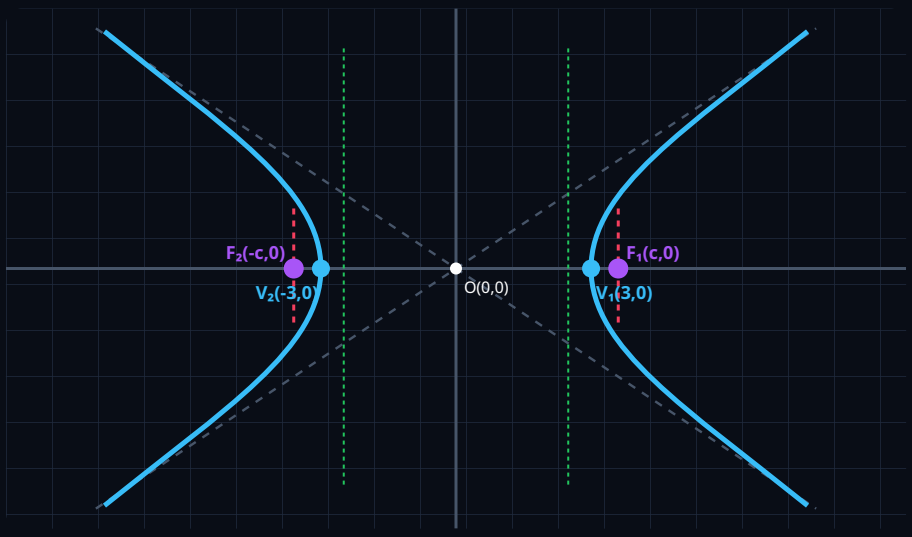
## 1. Definition & Fundamental Property

A **hyperbola** is the locus of all points in a plane such that the difference of their distances from two fixed points (called **foci**) is always constant.

$$| PF_1 - PF_2 | = 2a$$

where $F_1$ and $F_2$ are the foci, and $2a$ is the length of the transverse axis.

### Interactive Visual Representation (a = 3.0, b = 2.0)




Semi-Transverse Axis (a) 3.0

Semi-Conjugate Axis (b) 2.0

| FOCAL DIST. (C) | ECCENTRICITY (E) | LATUS RECTUM | VERTICES |
|---|---|---|---|
| 3.61 | 1.20 | 2.67 | (±3, 0) |

## 2. Standard Equations & Formula Comparison

| Property | Horizontal Transverse Axis | Vertical Transverse Axis |
|---|---|---|
| **Standard Equation** | $(x^2 / a^2) - (y^2 / b^2) = 1$ | $(y^2 / a^2) - (x^2 / b^2) = 1$ |
| **Transverse Axis** | Along X-axis (Length = $2a$) | Along Y-axis (Length = $2a$) |
| **Conjugate Axis** | Along Y-axis (Length = $2b$) | Along X-axis (Length = $2b$) |
| **Foci Coordinates** | $(\pm c, 0)$ where $c = \sqrt{(a^2 + b^2)}$ | $(0, \pm c)$ where $c = \sqrt{(a^2 + b^2)}$ |
| **Vertices** | $(\pm a, 0)$ | $(0, \pm a)$ |
| **Eccentricity ($e$)** | $e = c / a = \sqrt{(1 + b^2/a^2)} > 1$ | $e = c / a = \sqrt{(1 + b^2/a^2)} > 1$ |

| Property | Horizontal Transverse Axis | Vertical Transverse Axis |
|---|---|---|
| Latus Rectum Length | $2b^2 / a$ | $2b^2 / a$ |
| Equations of Directrices | $x = \pm a / e$ | $y = \pm a / e$ |
| Asymptotes | $y = \pm (b / a) x$ | $x = \pm (b / a) y$ |

**Key NCERT Tips:**

- For a hyperbola, $c > a$, which implies $e > 1$ always.
- The positive term in the standard equation determines the transverse axis.